

Abhishek Janagoudar

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📄 Visa Valid till Feb 2027 ●● Available to Join Immediately on-site



Profile

Automation and Robotics Master's student with hands-on experience in ROS 2, SLAM, and LiDAR-based perception. Skilled in building autonomous mobile robot systems, digital twins (AAS), and PLC-based safety systems. Strong programming skills in Python and C++. Looking for working student or internship roles in robotics, automation, and intelligent systems.

Professional Experience

BÄR Automation GmbH

Mobile Robots Development Intern (Full-time Mandatory)

11/2025 – 04/2026
Stuttgart, Germany

- Built a ROS 2-based SLAM pipeline (Cartographer, AMCL, SLAM Toolbox) for autonomous mobile robots, reducing rotational drift (8deg/360deg) and improving mapping accuracy
- Designed a ROS bag-based offline testing workflow, reducing parameter tuning time and enabling 5–10x faster validation
- Created an automated digital twin generation system using Asset Administration Shell (AAS v3.1), standardizing 160+ asset attributes
- Integrated AAS pipelines aligned with VDA 5050, enabling interoperability between AGVs and industrial IoT platforms

J. Ray McDermott Engineering Services Private Ltd

Associate System Engineer - AVEVA Electrical Admin V12

02/2024 – 08/2024
Chennai, India

- **Configured** data-driven workflows in AVEVA Electrical, enabling automated generation of 50+ datasheets and outputs
- **Structured** data models to improve consistency and reduce manual design effort in EPCIC projects

Infosys Ltd.

Senior System Engineer - AVEVA Engineering V15 Electrical & Instrumentation Developer

09/2021 – 02/2024
Mysore, India

- Managed AVEVA Engineering platform for electrical and instrumentation projects, including system configuration and database maintenance
- Delivered 50+ engineering outputs (load lists, cable schedules, datasheets, drawings) using data-driven workflows
- Led Agile Scrum activities and coordinated sprint execution, and mentored 9 engineers
- Produced technical documentation for systems, data models, and workflows

Arkay Energy Ltd.

Electrical Operations and Maintenance Intern

07/2019
TN, India

Supported substation commissioning, O&M tasks, and industrial safety practices.

LARSEN AND TOUBRO ELECTRICALS

Electrical Commissioning Intern

06/2019
Khargone, India

Assisted in electrical system setup, testing, and troubleshooting.

Education

Master of Science (M.Sc) | Electrical and Information Technology

Darmstadt University of Applied Sciences

09/2024 – present
Darmstadt,
Germany

Modules: Industrial Robotics, Safety in Industrial Automation, Computer Vision, Industry 4.0 / IIoT, Advanced Programming Techniques, Human Machine Interface

Bachelor of Engineering (B.E) | Electrical and Electronics

The National Institute of Engineering

07/2017 – 09/2021
Mysore, India

SKILLS

- **Robotics:** ROS 2, SLAM (SLAM Toolbox, Cartographer, AMCL), Navigation 2 Stack, VDA 5050, Robot Kinematics, Gazebo, RViz
- **Computer Vision:** Object Detection & Segmentation, Multi-Object Tracking, Camera Calibration, LiDAR & Point Cloud Processing, OpenCV
- **IIoT & Digital Systems:** OPC UA, MQTT, Node-RED, Digital Twins (AAS v3.1), Virtual Commissioning, IoT Gateway Integration
- **Automation & PLC:** TIA Portal, Siemens S7 Safety PLC, SISTEMA, EN ISO 13849-1
- **Programming:** Python, C++
- **AI & Automation:** Prompt Engineering, Ruflo (AI Agent Orchestration), Workflow Automation, Antigravity
- **Development Tools:** Visual Studio Code, Git, Linux, Docker
- **AI Tools & Concepts:** LLMs, AI-assisted coding tools (Cursor, Copilot), API-based model usage
- **Engineering Tools:** AVEVA Engineering v15.x, AVEVA Electrical v12, Jira, Rally
- **Soft Skills:** Agile/Scrum, Technical Training & Mentoring, Cross-team Coordination, CI/CD

Projects

BAER AAS Tools – Digital Twin Platform (ARENA2036): Developed digital twins of AGVs using Asset Administration Shell (AAS v3.1) for static and dynamic assets

- Built a Python-based generator to create and validate 160+ IDTA-compliant attributes, eliminating manual errors
- Automated digital twin creation and integration with the Eclipse BaSyx registry using REST APIs
- Ensured interoperability with VDA 5050 for AGV fleet communication
- Designed a web-based dashboard for monitoring asset status and synchronization
- Enabled export of simulation environments (Gazebo .sdf) for virtual testing
- **Published project repository:** github.com/abhishekrjanagoudar/aas_arena

LiDAR Mapping & Indoor Scene Analysis

Implemented ROS 2 pipeline for LiDAR mapping, plane segmentation, and environment reconstruction

UR5 / UR10 Kinematics

Modelled robot kinematics in MATLAB/Simulink

Advanced Programming Technique Lab Projects

- Developed console-based games (Battleship and Reversi) and a university enrollment database system in C++, applying OOP, game logic, and data management.

Safety in Industrial Automation Lab

- Programmed Siemens S7-1516F safety PLC and ET200SP modules in TIA Portal, implementing safety logic (ESTOP1, TWO_H_EN, SFDOOR, FDBACK) and validated compliance with EN ISO 13849-1 using SISTEMA (PL verification, black box testing).

Languages

English IELTS - 7.5 (C1)	Deutsch A2.2	Hindi Native	Kannada Native
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